

UVA DS2003: Communicating with Data

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Intro to Pandas and Data Frames

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Adapted from Hannah Bako and Rafael Alvarado

Anatomy of a Data Frame

ANATOMY OF PANDAS DATA STRUCTURES

A **Pandas DataFrame** is built from parts:

row names (axis 0)

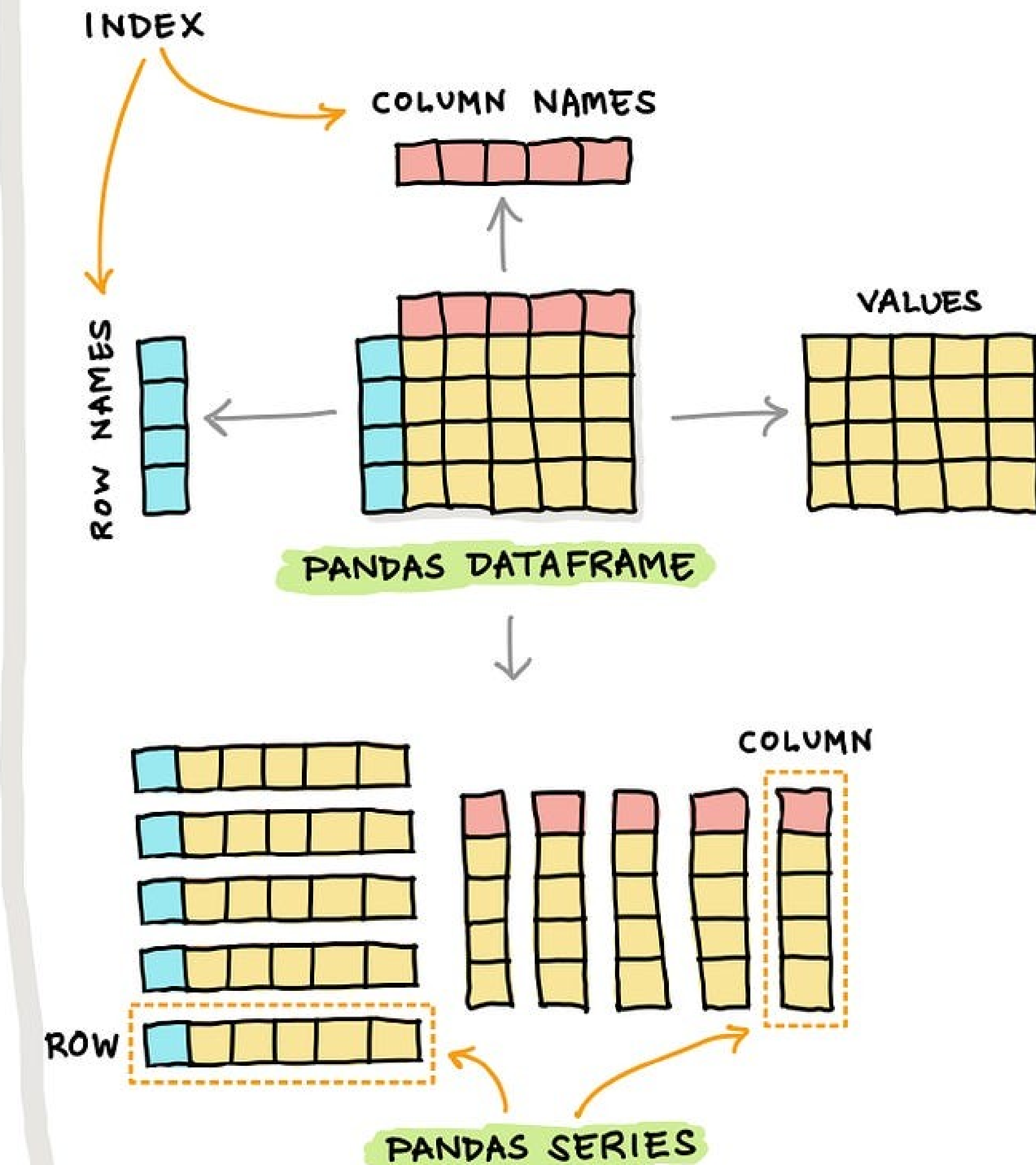
column names (axis 1)

Values

They can be processed in two ways:

row-wise

column-wise



BTW: Pandas, NumPy, and Matplotlib

Standard **data analysis** tool in Python

Not the only **game** in town

See Xarray, Polars, etc.

Builds on top of **NumPy** and **Matplotlib**

NumPy

Numerical Python; **array**-oriented programming

Matplotlib

Visualization package

In Pandas,
DataFrame objects
are actually built out
of **Series** objects

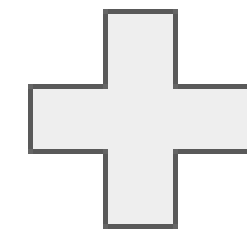
The diagram illustrates the structure of a Pandas Series. It features a table with three columns and seven rows. The first column contains index numbers (0-5), the second contains index labels (A-F), and the third contains the data values (5.6, 1.1, 2.6, 10.0, 9.5, 3.4). The column headers are 'weight' and 'weight'. Red arrows point from text labels to specific parts of the table: 'the index number' points to the first column header, 'the index labels' points to the second column header, 'the Series name' points to the 'weight' header, 'the index' points to the first column of data, and 'the data' points to the third column of data. A blue border highlights the first two columns, and a green border highlights the third column.

		weight
0	A	5.6
1	B	1.1
2	C	2.6
3	D	10.0
4	E	9.5
5	F	3.4

Anatomy of a Series

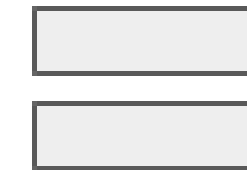
Series 1

		weight
0	A	5.6
1	B	1.1
2	C	2.6
3	D	10.0
4	E	9.5
5	F	3.4



Series 2

		height
0	A	54
1	B	22
2	C	34
3	D	98
4	E	80
5	F	42



DataFrame

		weight	height
0	A	5.6	54
1	B	1.1	22
2	C	2.6	34
3	D	10.0	98
4	E	9.5	80
5	F	3.4	42

A DataFrame is actually **made from** Series objects that **share an index**

SCALE	COMMON FUNCTIONS	COMMON PLOTS	
nominal	<code>df.value_counts()</code> <code>pd.crosstab()</code> <code>df[col].astype('category')</code>	Bar, Pie, crosstabs	
ordinal	<code>df.value_counts()</code> <code>pd.crosstab()</code> <code>df[col].astype('category')</code>	Bar, Line, crosstabs	sequential x-axis shows trend for Bar and Line
interval	<code>pd.to_datetime()</code>	Line, Histogram, Box, Scatter	
ratio		Line, Histogram, Box, Scatter	
nominal +	<code>df.groupby()</code>	Line, Histogram, Box, Scatter	nominal date is often used for grouping